

Module 1

Signal & Systems

Exercise Sheet

Section A – Basic Theory

Problem 1

Define the following terms:

- Signal
- System
- Continuous-time signal
- Discrete-time signal

Problem 2

Differentiate between:

- Continuous-time and discrete-time signals.
- Analog and digital signals.

Problem 3

Classify the following signals as:

- Continuous or Discrete
- Deterministic or Random
- Periodic or Aperiodic

(a) $x(t) = \sin(4t)$

(b) $x[n] = (-1)^n$

(c) Daily stock prices

(d) ECG signal

Problem 4

Determine whether the following systems are:

- Linear / Nonlinear
- Time-invariant / Time-varying

(a) $y(t) = 3x(t) + 5$

(b) $y(t) = x(t-2)$

(c) $y(t) = tx(t)$

(d) $y[n] = 2x[n]$

Problem 5

Determine whether the following systems are:

- Causal / Non-causal
- Memoryless / With memory

(a) $y(t) = x(t)$

(b) $y(t) = x(t+1)$

(c) $y[n] = x[n-2]$

(d) $y[n] = \int_{-\infty}^t x(\tau) d\tau$

Section B – Numerical Problems

Problem 6

Find the energy and power of

$$x(t) = 3, \quad 0 \leq t \leq 2$$

Problem 7

Find the energy of

$$x(t) = e^{-t} u(t)$$

Problem 8

Find the average power of

$$x(t) = 5\cos(2t)$$

Problem 9

Determine whether the following signals are energy or power signals.

(a) $e^{-2t} u(t)$

(b) $5 \cos(3t)$

(c) $\delta(t)$

Problem 10

Find the even and odd components of

$$x(t) = t^3 + t$$

Section C – Python Exercises

Problem 11

Generate and plot:

- Unit impulse
- Unit step
- Ramp signal

Problem 12

Generate and plot

$$x(t) = \sin(2\pi t) \quad \text{for } 0 \leq t \leq 2.$$

Problem 13

Generate and plot

$$x(t) = e^{-t} \quad \text{for } 0 \leq t \leq 5.$$

Problem 14

Generate and compare $\sin(2t)$ and $2\sin(2t)$ on the same graph.

Problem 15

Write a Python program to determine the energy of

$$x(t) = e^{-t} u(t)$$

using numerical integration.